

Varsha S

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SKILLS SUMMARY

- **Languages:** Python, Java, JavaScript, SQL, HTML, CSS
- **Frameworks & Libraries:** Flask, Django, FastAPI, React.js, Node.js, Express.js, Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, Keras, OpenCV, Seaborn, Matplotlib
- **Tools & Technologies:** Git, Docker, Kafka, Redis, Postman, Power BI, Tableau, Excel, MySQL, MongoDB, PostgreSQL, Firebase, AWS, Google Cloud Platform, Render
- **Platforms & IDEs:** Jupyter Notebook, Google Colab, PyCharm, Visual Studio Code, IntelliJ IDEA
- **Core Concepts:** Data Structures & Algorithms, OOP, REST APIs, Microservices, System Design Basics, Data Visualization, Statistical Modeling, MLOps, Cloud Deployment
- **Soft Skills:** Analytical Thinking, Problem Solving, Collaboration, Stakeholder Management, Communication, Leadership, Adaptability

WORK EXPERIENCE

DATA SCIENCE INTERN | AMILO AI PVT LTD

February 25 – May 25

- Developed and implemented **web scraping pipelines** to extract data from various social media platforms.
- Pre-processed and structured large-scale datasets to enhance model training efficiency.
- Trained machine learning models to generate **AI-driven content scripts** for trending topics.
- Integrated **text-to-video AI models**, automating script-to-video conversion for content creators.
- Collaborated with cross-functional teams to improve AI model performance and user experience.

Virtual Internships (Forage)

- **J.P. Morgan Chase (Jan 2025):** Set up Spring Boot + Kafka, built REST APIs for data streaming.
- **Accenture (Jan 2025):** Cleaned and modeled data with Python & SQL; visualized insights in Power BI.
- **TATA Group (Nov 2024):** Designed IAM solutions to strengthen enterprise security.

PROJECTS

1. AI-Powered Academic Project Evaluation System-Python | Flask | BERT | ResNet-50 | I3D | MongoDB

- Developed a **multimodal AI evaluation platform** integrating text, image, and video analysis to assess academic projects.
- Implemented **BERT-based NLP model** for domain understanding and **ResNet-50 + I3D** for visual feature extraction.
- Achieved **90% evaluation accuracy**, reducing manual grading time by **80%** and ensuring unbiased assessment.
- Deployed a scalable Flask web application with **MongoDB cloud storage** and **Firebase authentication**.

2. Medical Image Classifier-CNN | TensorFlow | Transfer Learning | Keras

- Designed a deep learning model based on **ResNet architecture** to classify X-ray and MRI images for disease detection.
- Applied **transfer learning** for higher accuracy and better generalization across multiple medical datasets.
- Achieved **94% classification accuracy** and visualized model performance using **Grad-CAM** heatmaps.

3. Sentiment Analyzer – Social Media Data-Python | NLP | Transformers | Flask

- Built an **NLP-based sentiment analysis model** leveraging **BERT transformer architecture** for contextual understanding.
- Processed large-scale real-world text data from social media to classify sentiments into multiple categories.
- Achieved **90% precision**, improving over baseline models using fine-tuned hyperparameters and preprocessing pipelines.
- Integrated the model with a **Flask API** for real-time sentiment prediction on user-entered text.

4. AI Memory Palace- Python | Generative AI | LLMs | Flask

- Created an **interactive AI-powered memory assistant** using **Large Language Models (LLMs)** for personalized knowledge recall.
- Implemented semantic embeddings to enable **context retention and conversation continuity**.
- Deployed via a **Flask web interface** with OpenAI API integration for dynamic knowledge retrieval.

5. **Smart Inventory System**-*TypeScript | Node.js | Express.js | ML API | React.js | MongoDB*
 - Developed a **full-stack inventory management web app** with AI-driven demand forecasting and real-time stock tracking.
 - Integrated **RESTful APIs**, **JWT authentication**, and a machine learning module for predicting product restock cycles.
 - Built a responsive **React.js frontend** and connected to **MongoDB** for scalable cloud-based data handling.
6. **Yoga Pose Detection & Correction**-*OpenCV | Mediapipe | TensorFlow*
 - Implemented a **real-time yoga posture analysis system** using **body landmark detection**.
 - Achieved **90% accuracy** across 15+ yoga poses with Mediapipe’s pose estimation pipeline.
 - Provided real-time feedback and correction guidance through live camera input.
7. **Customer Segmentation and Market Insights** - *K-Means Clustering | EDA | Python | Seaborn | Matplotlib*
 - Conducted **exploratory data analysis (EDA)** and applied **K-Means clustering** to segment customers based on behavior and spending.
 - Created **data visualizations** and **insightful dashboards** to support targeted marketing strategies.
 - Delivered actionable recommendations improving campaign personalization and user retention.
8. **Multi-Camera Real-Time Human Activity Recognition System** – *OpenCV, YOLOv8-Pose, LSTM/Transformers, WebSockets*
 - Developed a real-time system for multi-camera person tracking and action recognition (falls, exercises, anomalies) with a live analytics dashboard.
9. **AI Research Assistant with RAG & Knowledge Graphs** – *LangChain, LLMs, Pinecone, FastAPI*
 - Engineered a research assistant that extracts insights from PDFs, builds semantic search, and generates knowledge graphs for cross-paper analysis.
10. **Autonomous Social Media Marketing AI Agent** – *LLM Agents, LangGraph, DALL-E/SDXL, Instagram API, Next.js*
 - Built an end-to-end AI agent that **creates content**, designs images, predicts engagement, posts automatically, and adapts strategy using analytics.
- CERTIFICATES
- Artificial Intelligence Fundamentals – *IBM*, Nov 2024

Generative AI Certified Professional (OCI) – *Oracle*, Jul 2024

Data Science Professional(OCI)-*Oracle*,Oct 2025

Cybersecurity Analyst Job Simulation – *Tata Group (Forage)*

Data Analytics and Visualization Job Simulation – *Accenture North America (Forage)*

Software Engineering Job Simulation – *J.P. Morgan (Forage)*
- EDUCATION
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|--|----------------|
| Atria Institute of Technology – Bengaluru, India | 2022 – Present |
| Bachelor of Engineering (CSE/Data Science) | |
| PES Pu College – Bengaluru, India | 2020-2022 |
| BGS World School– Bengaluru, India | 2010-2020 |
- EXTRA-CURRICULAR ACTIVITIES
- Hacktoberfest Contributor
 - Participated in Hacktoberfest, contributing to open-source projects by fixing bugs and adding features.

AryaX AI Hackathon Contributor
 - Collaborated with a team on AI-focused projects as part of the AryaX AI hackathon. Developed innovative AI solutions and presented findings in competitive settings

IEEE Runtime 24
 - Participated in the IEEE Runtime 24 Hackathon and ranked in top 10 teams out of 100+ teams